PlanArena: an exact, learning-assisted land-use optimizer with empirical evaluation

Abstract

Deciding which land uses go where is a location-allocation problem with multidimensional effects: a change alters homes, jobs, access to services, the tax base, emissions and runoff, interacts with neighbouring parcels, and may never be built. PlanArena is an open land-use optimizer on official data for 4,995,208 parcels in ten US cities. It scores any parcel-level change on 11 indicators in 46 ms (Manhattan) to 0.58 s (New York City), estimates how likely a change is, finds the best plan within zoning and precedent, and compares nine voting rules over simulated stakeholders. A mixed-integer formulation of the score certifies the optimum: it matched exhaustive search on 20 blocks and proved the optimal plan of 29 of 30 areas of up to 411 parcels (median 3.2 s) and of nine of 12 districts. A graph neural network trained on certified optima proposes plans in 80 ms, as close to the optimum as first-order search. On held-out Manhattan lots, given what was built, predicted changes correlated with recorded homes ($r = 0.79$) and emissions ($r = 0.60$) but overstated total homes 1.47-fold and had no skill for jobs; a learned correction cut the error in homes by 15%, and conformal intervals covered 85%–98% of changes, against 19%–66% for sensitivity ranges. Redevelopment was ranked out of time with an area under the receiver operating characteristic curve (AUC) of 0.851–0.902, and likelihood weighting brought housing projections from 76–193 to 0.80–2.60 times what was built. The votes are illustrative.

Keywords: Land-use optimization · Mixed-integer programming · Graph neural networks · Urban digital twin · Planning support systems · Model validation

1 Introduction

Deciding which land uses go where (housing, jobs, services, open space) is a location-allocation problem (Church & ReVelle, 1974; ReVelle & Eiselt, 2005). Its effects are hard to measure for four reasons.

- **Many dimensions.** One change alters homes, jobs, access to shops, parks and facilities, the tax base, energy use and runoff, and these trade off against each other (Malczewski, 2006; Rahman & Szabó, 2021).
- **Spatial interaction.** Effects spread to neighbouring parcels through walking distance, land-use mix and competition for development (Verburg et al., 2004).
- **Uncertainty.** A permitted change may never be built. In three Manhattan rezonings replayed here, capacity projected 8,423 homes on development sites; 561 were built by 2025.
- **Competing interests.** Residents, developers, businesses, environmental advocates and city finances weigh these effects differently, and no single answer is correct (Rittel & Webber, 1973).

How decisions are made today. In US cities, rezonings and plan amendments are decided at public hearings.

- **The process.** Planning staff prepare a report, and large rezonings, such as those replayed here, add an environmental review that projects development on identified sites. Residents and advocates testify; the planning commission and the council vote.
- **Testimony is advisory.** Residents are heard but do not vote. Speakers at planning and zoning meetings are disproportionately older, male, long-time homeowners (Einstein et al., 2019), and legally required hearings meet few goals of participation (Arnstein, 1969; Innes & Booher, 2004; Fung, 2006).
- **Fragmented evidence.** Environmental, fiscal and access effects arrive as separate studies, if at all, and are rarely weighed together.

What computation can measure. Each part of the problem now has a technical tool:

- **parcel-level impact models** calibrated on observed records, which measure many indicators at once;
- **supervised learning on parcel histories**, tested out of time, which estimates how likely a change is (Tepe & Guldmann, 2020; Carruthers & Wei, 2024);
- **graph neural networks** over parcel adjacency, which represent spatial dependence;
- **mixed-integer programming**, which can prove a land-use allocation optimal (Aerts et al., 2003), and machine learning that speeds exact search without losing its guarantee (Gasse et al., 2019; Nair et al., 2020);
- **many-objective evolutionary search** (Deb & Jain, 2014), which explores the combinatorial space of plans;

- **computational social choice and bargaining** (Nash, 1950; Brandt et al., 2016), which turn stakeholder preferences into a decision. Game-theoretic simulation has been applied to collective land-use decisions (Abolhasani et al., 2022).

Planning support systems (PSS) and urban digital twins aim to combine such tools (Klosterman, 1999; Geertman & Stillwell, 2020; Batty, 2018; Lehtola et al., 2022). Three gaps remain:

- **Untested outputs.** Their outputs are seldom tested against outcomes (Vonk et al., 2005; Lei et al., 2023; Azadi et al., 2025; Kruse et al., 2026).
- **Optimization.** It rarely asks whether recommended changes are optimal or plausible, or how the choice depends on the decision rule (Zheng et al., 2023; Feng et al., 2024; Lin et al., 2025).
- **Fragmented data.** The data are spread across agencies and schemas, some only as documents: Philadelphia's land-use plan exists only as maps in 17 PDF district plans.

This paper. PlanArena is an open land-use optimizer that measures the impact of any parcel-level change and makes stakeholder interests part of the decision. It contributes:

1. **Measurement.** Any edit is scored on 11 indicators in three pillars, with sensitivity ranges, at interactive speed. Each edit also shows what exists, what zoning permits, what the plan intends and how likely the change is, and a learned correction of homes and value with calibrated intervals.
2. **Optimization and decision.** The optimizer finds the plan that maximizes the score within zoning and precedent and certifies it: optimal for areas of up to a few hundred parcels and for most districts, with a proven gap otherwise. Exact trade-off fronts show what each pillar costs in the others and how far today's land use lies inside the frontier, and a graph neural network trained on certified optima proposes plans at once. Resident, environmental, business, developer and fiscal interests vote on candidate plans under nine rules, and residents also vote block by block.
3. **Official data for ten cities.** Existing use, plan and zoning for 4,995,208 parcels are on one schema, with a validated method to digitise plans published as vector PDF maps.
4. **Empirical evaluation where data allow** (Sect. 4): held-out and out-of-time tests of impacts, likelihood and projections, exhaustive checks and certificates of the searches, and the type of evidence behind each claim, with design guidance drawn from the results.

Scope. Impacts are evaluated in Manhattan only; elsewhere coefficients are national or regional. Zoning is represented by use permissions and maximum floor-area ratios, access by straight-line distance. Votes rest on assumed utilities, not elicited preferences, and usability with planners is not evaluated. Online Resource 1 specifies every model and test.

2 Related work

PSS and urban digital twins. PSS combine suitability, demand projection and allocation for scenario planning (Klosterman, 1999; Geertman & Stillwell, 2020), often in a geodesign workflow linking representation, change, impact and decision (Steinitz, 2012). Their limited use and uncertain usefulness are long-standing concerns (Vonk et al., 2005; te Brömmelstroet, 2013; Pelzer, 2017). Digital twins extend them to integrated, updatable city models (Batty, 2018, 2024; White et al., 2021; Lehtola et al., 2022; Ferré-Bigorra et al., 2022). Reviews report technical and organisational barriers (Lei et al., 2023; Wu & Guan, 2025), and ambitions ahead of demonstrated contributions (Azadi et al., 2025; Luo et al., 2025; Kruse et al., 2026). PlanArena attaches a test against observed data to each output.

Optimization and collective decisions. Location-allocation models site facilities and uses (Church & ReVelle, 1974; ReVelle & Eiselt, 2005); multicriteria analysis and multi-objective optimization generate and rank land-use alternatives (Malczewski, 2006; Rahman & Szabó, 2021); integer programming solves allocation problems to proven optimality (Aerts et al., 2003), and efficiency frontiers show what one objective costs in another (Polasky et al., 2008). Recent work uses deep reinforcement learning (Zheng et al., 2023), which offers no certificate, and explicit stakeholder interests (Feng et al., 2024; Lin et al., 2025). In exact optimization, learned models choose branches or propose partial solutions while the solver keeps its guarantee (Gasse et al., 2019; Nair et al., 2020). Planning problems have no single correct solution (Rittel & Webber, 1973), so the rule that turns preferences into a choice matters. Bargaining (Nash, 1950) and voting rules (Schulze, 2011; Brandt et al., 2016) have different properties, and bargaining has been combined with land reallocation (Hashemi et al., 2026). Simulated votes are not participation (Arnstein, 1969); PlanArena uses them only to show how a choice depends on its rule.

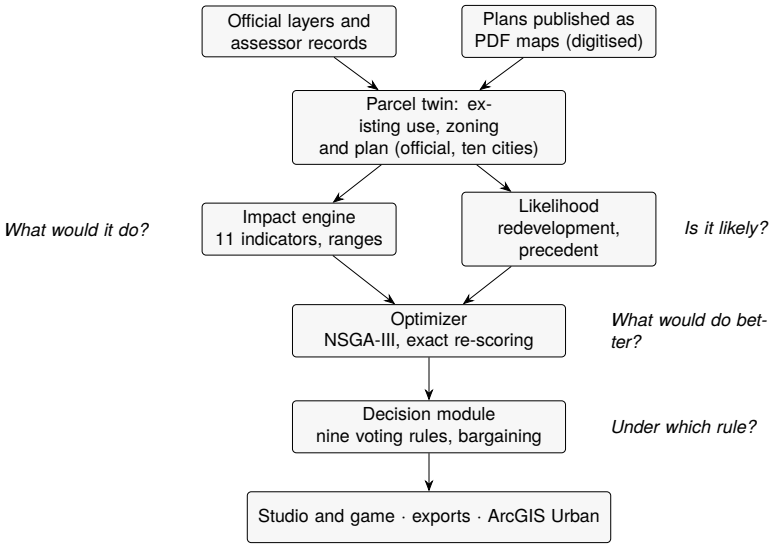


Fig. 1 Architecture. Every number the system shows comes from one of these components, each with a stated test (Sect. 4)

Predicting development, and regulation. Urban change models range from cellular automata (Clarke et al., 1997) and statistical site models (Landis & Zhang, 1998) to behavioural (Waddell, 2002; Waddell et al., 2003) and agent-based simulation (Parker et al., 2003), with neighbourhood and temporal effects as priorities (Verburg et al., 2004; Weerasinghe et al., 2025).

- **Parcel models** capture existing and historical neighbourhood conditions (Tepe & Guldmann, 2020), and machine learning complements econometrics (Carruthers & Wei, 2024).
- **Redevelopment** depends on land value relative to the structure (Rosenthal & Helsley, 1994; Dye & McMillen, 2007). In Chicago, smaller and older buildings with less lot coverage are replaced more often (Weber et al., 2006; Charles, 2013).
- **Regulation is not exogenous.** Rezoning follows development opportunity (Gabbe, 2018), studies differ on upzoning's short-run effect on construction (Freemark, 2020; Dong, 2024; Greenaway-McGrevy & Phillips, 2023), and plans are only partly implemented (Talen, 1996; Laurian et al., 2004; Brody & Highfield, 2005; Berke et al., 2006; Loh, 2011).
- **Regulation in models.** Recent land-change models integrate zoning and planning rules (Hersperger et al., 2018; Domingo et al., 2021; Abolhasani et al., 2026) but do not test what they add to prediction on held-out history.

Validation. Land-change predictions should be compared with persistence at several resolutions (Pontius et al., 2008; Pontius & Millones, 2011), and their transition probabilities validated directly (Black et al., 2023). Under spatial and temporal dependence, random cross-validation overstates skill (Roberts et al., 2017; Ploton et al., 2020), and models apply only where new data resemble the training data (Meyer & Pebesma, 2021). Uncertainty from many coefficients is better assessed by varying them jointly (Saltelli et al., 2019).

3 The system

A parcel twin holds each city's official layers (Figure 1). Four engines answer the planner's questions:

- the **impact engine**: what would a change do?
- the **likelihood models**: is it likely?
- the **optimizer**: what would do better?
- the **decision module**: who would choose it, and under which rule?

In the Studio, a browser application, a planner paints uses and floor-area ratios on parcels and sees every indicator, flag and recommendation update (Figure 2).

Parcel twin and official layers. The twin covers the nine largest US cities except San Jose, plus Milwaukee (Table 1).

- **Existing land use** is the parcel land use the city or its regional planning agency publishes, refined with assessor records where the layer groups uses.

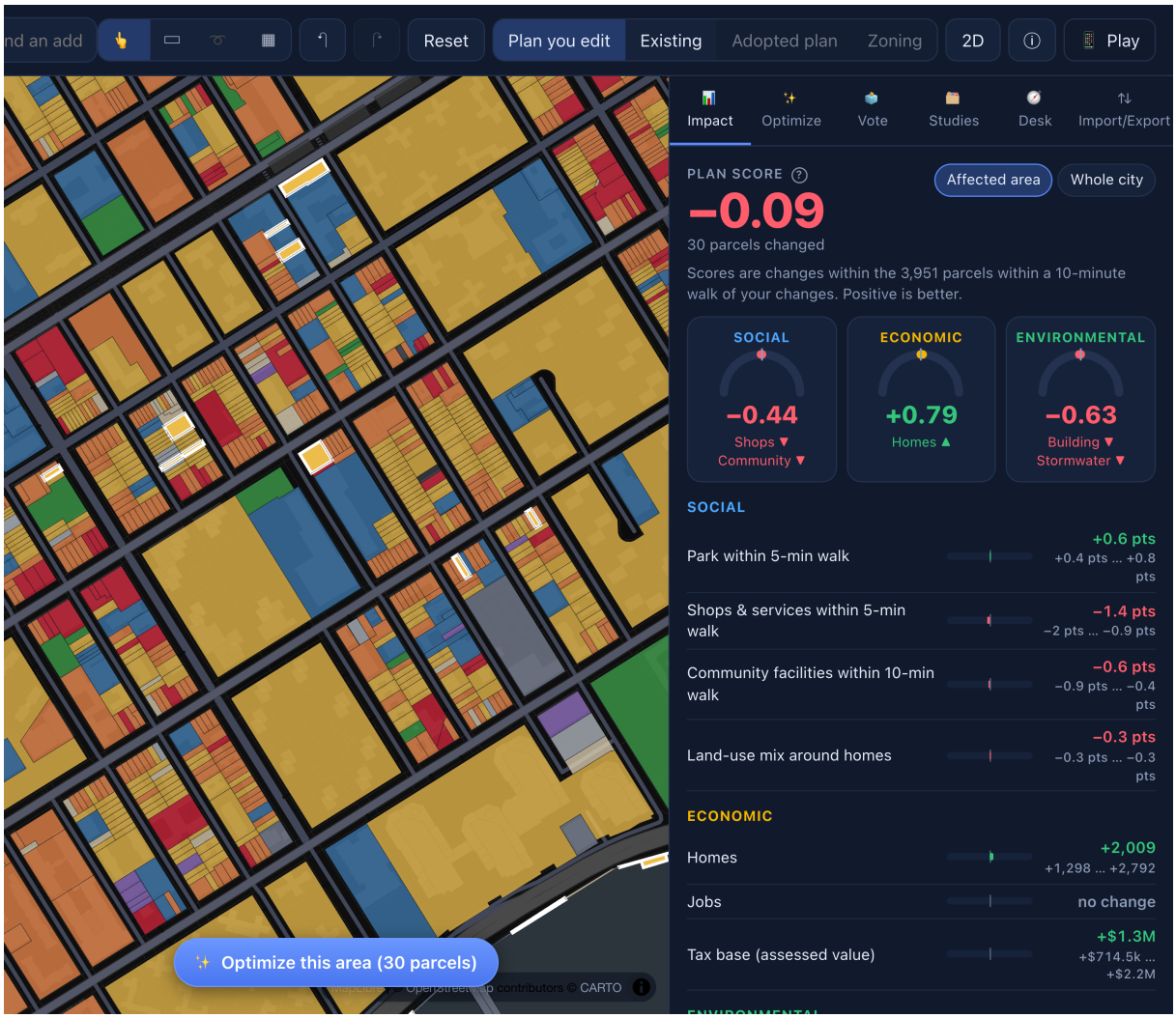


Fig. 2 The Studio in Manhattan: thirty parking lots and vacant parcels in East Harlem painted as housing (outlined). Right: plan score, social, economic and environmental pillars, and each indicator's change within a ten-minute walk with its sensitivity range. Top: switches between the plan being edited and the city's existing, planned and zoned use. The use palette, left of the map, is not shown

- **The adopted plan** (future land use) and **zoning** are separate layers.
- **Classes.** Explicit crosswalks map every source code to 16 classes: three housing types, mixed use, retail, office, hotel, industry, warehousing, education, health, civic, open space, transportation, parking and vacant land. The classes nest in nine analysis groups.
- **Records.** Floor area, homes, year built and assessed value come from assessor or city records, with estimated fields flagged. Jobs are fitted to block counts of the Longitudinal Employer–Household Dynamics (LEHD) programme (U.S. Census Bureau, n.d.).
- **Provenance.** Inputs are downloaded by script, with URL and SHA-256 hash recorded.
- **Philadelphia's plan**, published only as PDF maps, is digitised (Appendix A): 15 of 17 plans (Philadelphia City Planning Commission, n.d.), covering 83% of parcels, with median registration errors of 0.3–2.2 m, each checked against a shifted-map control.
- **Missing plans.** Where a city publishes no plan as data, the layer is absent, not estimated.

Impact engine. An edit sets a parcel's use and, optionally, its floor-area ratio (FAR). The engine computes the 11 indicators of Table 2 within a ten-minute walk and citywide.

- **Unchanged parcels** keep their observed values, so the baseline reproduces the records exactly.
- **Changed parcels** follow a hybrid delta rule. A density-only change on a consistent record scales the observed composition; a use change removes the observed old use and adds the modelled new one (Online Resource 1, Sect. S3).
- **Derived quantities.** Homes follow from residential floor area and the parcel's floor area per home. Jobs follow from floor-area components and rates fitted to LEHD counts. Emissions follow from energy intensities and

Table 1 The ten cities: official parcel layers on one schema. Shares are of the city's parcels.

City	Parcels	Existing use (source)	Share	Adopted plan	Share	Zoning	Homes / Census
New York	856,634	MapPLUTO (City Planning)	100%	none citywide	–	100%	1.03
Los Angeles	672,444	SCAG 2020; county assessor	100%	General Plan (SCAG 2019)	100%	100%	0.99
Chicago	606,894	CMAP 2023; county assessor	100%	none citywide	–	96%	1.05
Houston	630,525	City (appraisal codes)	97%	none	–	–	0.95
Phoenix	510,871	County assessor; MAG	100%	General Plan	100%	99%	0.99
Philadelphia	547,410	City Planning Commission; OPA	100%	district plans (digitised PDF)	83%	100%	0.99
San Antonio	438,009	Bexar Appraisal District	100%	adopted area plans	61%	82%	0.99
San Diego	272,948	SANDAG 2025; county assessor	100%	General and community plans	100%	94%	1.02
Dallas	311,431	City 2026; DCAD	100%	forwardDallas! 2.0	99%	79%	0.98
Milwaukee	148,042	MPROP	100%	PDF only (not included)	–	99%	1.06
All	4,995,208						

Existing use is the city's or regional agency's parcel land use, refined by assessor records where the layer groups uses. Homes / Census: homes in the twin over Census 2020 housing units. Houston has no zoning. Milwaukee's area plans are published as PDFs that could not be downloaded by script. Sources: New York (New York City Department of City Planning, 2025); Los Angeles (Southern California Association of Governments, n.d.); Chicago (Chicago Metropolitan Agency for Planning, n.d.); Houston (City of Houston, n.d.); Phoenix (Maricopa County Assessor, n.d.); Philadelphia (City of Philadelphia, n.d.); San Antonio (Bexar Appraisal District, 2025); San Diego (San Diego Association of Governments, 2025); Dallas (City of Dallas, n.d.); Milwaukee (City of Milwaukee, n.d.).

Table 2 Indicators computed for every edit, and how each is tested (Manhattan).

Indicator	Pillar	Unit	Better	Test
Homes	economic	units	higher	held-out change
Jobs	economic	jobs	higher	existing; change: no skill
Tax base (assessed value)	economic	USD	higher	held-out change
Park within 5-min walk	social	% of residents	higher	not validated
Shops & services within 5-min walk	social	adequacy 0-1	higher	not validated
Community facilities within 10-min walk	social	adequacy 0-1	higher	not validated
Land-use mix around homes	social	index 0-1	higher	not validated
Green & open space	environmental	ha	higher	direct accounting
Stormwater runoff	environmental	impervious-equivalent ha	lower	not validated
Building emissions	environmental	tCO2e / yr	lower	existing; held-out change
Green cover near homes	environmental	% of land within 400 m	higher	not validated

Unchanged parcels keep their observed values; changed parcels follow the delta equations. Each indicator carries a sensitivity range from low and high coefficient variants. Access, mix, green cover and runoff are recomputed from the resulting state but have no observed after-change data to test against. The Studio reports areas in acres (1 acre = 0.405 ha).

emission factors derived from New York's benchmarking data (Local Law 84, LL84) and emission limits (Local Law 97) (New York City Department of Buildings, 2023). Runoff follows from surface coefficients, and value from the difference between modelled values.

- **Access, mix and green cover** are recomputed on a 50 m raster within straight-line radii of 400 m and 800 m, not along walking routes.
- **Ranges.** Every coefficient has a source and a status, and low and high variants give each indicator a sensitivity range. A range has no probability attached and is not a prediction interval.
- **Learned correction.** For homes and value, a gradient-boosted model of the engine's error, trained on observed changes, and conformalized quantile regression (Romano et al., 2019) give a corrected yield and a 90% prediction interval for each parcel and use, shown beside the engine's in the parcel inspector; the engine's scores are unchanged (Sect. 5.2).

Indicator scores are per cent changes (totals) or percentage points (shares) against the scope's baseline, clipped to ± 100; a pillar is the mean of its indicators, and the overall score a weighted mean of pillars with weights the planner sets. Scoring a 50-parcel edit, ranges included, takes 46 ms on Manhattan's 42,075 parcels and 0.58 s on New York City's 856,634. Impacts are computed for the nine analysis groups, for which the coefficients are calibrated and validated.

Likelihood, precedent and pattern. Three models judge plausibility.

- **Redevelopment.** A gradient-boosted model (LightGBM), trained only on outcomes known at the decision year, gives each lot's probability of redevelopment within one, three or five years and its likely new use. Its inputs are the lot's use, size, built and permitted floor area, age and land share of value, its neighbourhood and its zoning.
- **Precedent.** A conversion is flagged if buildings of that use and size were not converted to the proposed use in the past decade.
- **Pattern.** A graph neural network over parcel adjacency, combined with LightGBM, describes how well a use fits its surroundings: an existing pattern, not a norm.

Edits that depart from the plan or from what zoning allows are flagged (Figure 3).



Fig. 3 Parcel inspector on the adopted-plan layer (Dallas). For a vacant lot the plan designates Community Mixed-Use, it lists existing use, plan designation (flagged as differing), zoning, the uses zoning allows, and what each new use would yield in homes, jobs and assessed value

Optimizer. The optimizer maximizes the overall score minus a penalty per conversion, under a budget of changes. The uses zoning permits are a hard constraint, existing homes and protected sites are kept, conversions without precedent are skipped, and candidates can be restricted to the parcels most likely to be redeveloped.

- **Fast search.** Iterated best response over first-order move estimates, with exact re-scoring of the best moves, gives a plan at once.
- **Exact search.** The score is stated as a mixed-integer linear program and solved by branch-and-bound (Huangfu & Hall, 2018). Totals are linear in the choices. The five resident-weighted shares become one weighted mean whose numerator and denominator are linear after exact products of binary choices and bounded quantities (McCormick, 1976). Park access within a walk is a covering constraint (Church & ReVelle, 1974), and land-use mix enters through tangents of the entropy. Every relaxation over-estimates the score, so the solver's bound is a certificate: no feasible plan scores higher. The plan is re-scored by the engine, and the model is tightened at the plan until bound and score agree (Online Resource 1, Sect. S6).
- **Trade-offs.** Optimal plans for a grid of pillar weights, the plan that maximizes the weakest pillar and the ε-constraint method (Mavrotas & Florios, 2013) give the efficiency frontier (Polasky et al., 2008), trade-off rates between pillars, and how far today's land use lies inside the frontier. NSGA-III (Deb & Jain, 2014) remains for fast exploration of a selection.
- **GNN policy.** A graph neural network (GraphSAGE; Hamilton et al. 2017) trained on certified optima proposes a plan in milliseconds while a planner drags the trade-off, and starts the exact search (Nair et al., 2020). Its plan is a preview; the plan reported is the exact one.

In the Studio a click returns the fast plan, then the certified one; dragging a handle on a triangle of pillar weights moves the map along the trade-off; and a lasso around any area returns its optimal plan. A divider swiped across

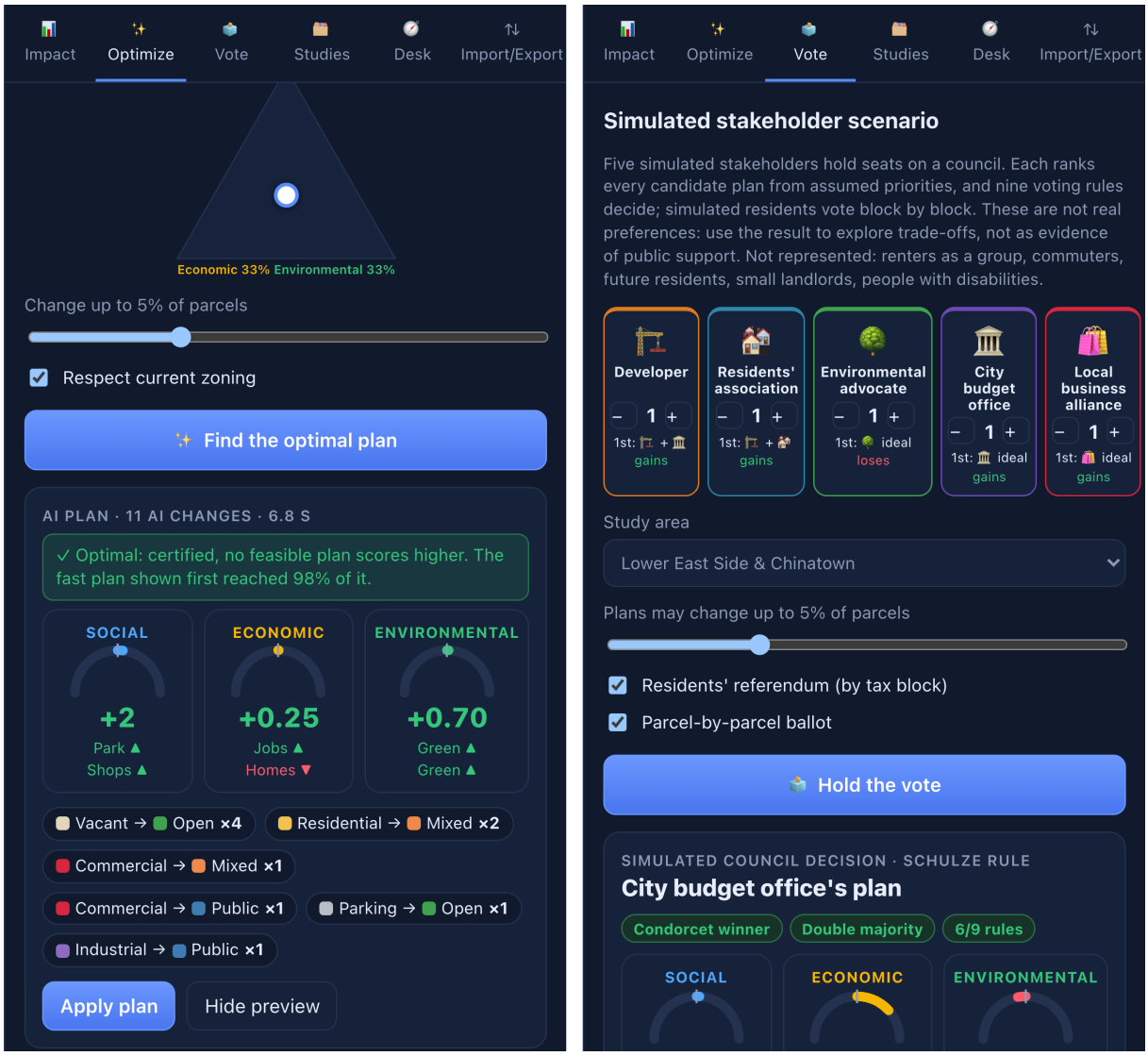


Fig. 4 Optimizer and decision module in Manhattan. Left: the certified optimal plan of an area in East Harlem selected on the map, the trade-off triangle, the plan's change in each pillar and its conversions. Right: the simulated council's decision under Schulze's rule, whether the winner is a Condorcet winner with a double majority, and how many of the nine rules it wins

the map compares today's land use with the plan.

Decision module. Choices are contested (Rittel & Webber, 1973), so the module shows how the chosen plan depends on the decision rule.

- **Agents.** Five stakeholder agents (residents, business, developers, environmental and city-finance interests) score candidates from assumed utilities over the indicators.
- **Rules.** Nine voting rules are compared with the Nash bargaining solution (Nash, 1950) and a residents' referendum by block. Schulze's method (Schulze, 2011) decides.
- **Reports.** Whether a Condorcet winner exists, whether one insincere ballot could change the outcome, and the winner's robustness to coefficients and seats.

Unlike hearing testimony, these interests are votes: each agent holds one seat on a simulated council, and residents also vote block by block. The agents stand for interests, not for people, and one agent per interest does not capture the variety within it. Utilities are editable assumptions, and the output is a simulated scenario, not public support (Arnstein, 1969) (Figure 4).

Interface. Planners can edit parcels by click, box, lasso or search, or upload plans as tables or GIS files, which are reviewed before they are applied. Studies are stored with a hash-chained audit log. Exports include a staff-report draft, CSV and GeoJSON files, and the provenance of every coefficient. A phone version presents the engine as a

Table 3 Evaluation design: each component, its question, data and type of evidence.

Component	Question	Data	Evidence	Section
Existing conditions	Does the twin reproduce what is there?	LL84 energy, LEHD jobs, recorded homes	accounting	5.1
Planning calculations	Does it reproduce a review's projection from the review's own inputs?	Environmental impact statements (two rezonings)	accounting	5.1
Effects of change	Given an observed change, does it predict the effect?	MapPLUTO 2015 vs 2025, held-out folds	empirical	5.2
Learned correction	Does learning lower the engine's error, and are its intervals calibrated?	The same held-out folds	empirical	5.2
Likelihood	Does it rank lots that redevelop, and how many, forward in time?	Rolling origins 2016–2025	empirical	5.3
Official layers	What do zoning and plans add, in other cities?	Seven city panels	empirical	5.4
Housing projections	Are projected homes close to what was built?	Four cities; three rezonings	empirical	5.5
Exact optimizer	Is the plan optimal, or how far can it be from the optimum?	Exhaustive search of blocks; areas; 12 districts	algorithmic	5.6
Trade-off front	Are front plans Pareto-optimal, and what does each pillar cost?	Exhaustive enumeration of blocks; areas	algorithmic	5.6
GNN policy	How close is an instant plan to the optimum?	Certified optima of held-out areas	empirical	5.6
Realism	Are recommended changes plausible?	2020 plans vs 2025 outcomes	empirical	5.6
Decision module	How does the choice depend on rule, seats and candidates?	Assumed utilities	illustrative	5.7

game. Any city or district exports to ArcGIS Urban: space-use, zoning and land-use types, boundaries, and parcels with a PlanArena scenario. An online demonstration of the Studio for Manhattan is at `https://planarena-manhattan.onrender.com` and the game at `https://planarena-manhattan.onrender.com/play` (an earlier release, v1.0.0-rc.1; the first load can take about a minute); the code for all ten cities is in the repository (Statements and Declarations).

4 Evaluation methods

The claims rest on four kinds of evidence (Table 3): *empirical predictive tests*, in which a prediction made with earlier information is compared with a later outcome on observations not used to build or select the model; *accounting checks*, in which the twin reproduces records it was built from; *algorithmic verification* of the searches against exhaustive enumeration; and *illustrative simulation* for the decision module, which has no observed counterpart.

- **Time.** Predictions are trained on one period and tested on a later one; uncertainty comes from resampling spatial tiles (Kunsch, 1989), never from random splits (Roberts et al., 2017). The cross-city test uses seven cities with parcel histories (Table S13 in Online Resource 1).
- **Development** is defined as each source records it: a new building or large floor-area increase (Los Angeles, Dallas, New York), a new building (Milwaukee), a change of use class between land-use inventories (Phoenix, San Diego), or a new-construction permit (Chicago). A building already on the starting record is never counted as new.
- **Blocks.** Four feature blocks are added in turn: existing conditions (E), context within 400 m (C), zoning (Z) and plan (P).
- **Leakage control.** Every feature family is dated against the decision year (Online Resource 1, Sect. S2). Where a current zoning or plan layer has dated changes, a conservative variant drops lots whose zoning or plan dates from the window's start or later; lots that merge or split are followed by their representative point.
- **Data.** Manhattan tests use the annual MapPLUTO releases (New York City Department of City Planning, 2025), LL84 benchmarking (New York City Department of Buildings, 2016, 2023) and LEHD job counts (U.S. Census Bureau, n.d.); the city panels use the sources of Table 1.

Use of AI tools. Claude (Anthropic), used through Claude Code, assisted in writing and testing the code, running the analyses, and drafting the text and Online Resource 1. The author designed the study and the system, reviewed the code, results and text, read the cited sources and is accountable for the content. Every reference was checked against the record its DOI or ISBN resolves to. No AI system generated data or results: every number is computed by the released code from the public inputs, and tests fail if a reported number differs from the result files.

Table 4 Effects of observed change on 2,036 held-out Manhattan lots, 2015–2025: given the 2015 state and the observed new use and floor area, predicted against recorded change (Pearson r; mean absolute error and mean bias in homes per lot; predicted over observed total).

Method	Homes r	MAE	Bias	Total	Value r	Total	Emis. r	Total
PlanArena (deployed form)	0.79	12.5	5.6	1.47	0.74	0.70	0.60	1.24
PlanArena (form chosen on selection lots)	0.79	12.7	6.1	1.51	0.74	0.70	0.60	1.24
× selection-lot scale	0.79	11.8	4.2	1.35	0.74	1.09	0.60	1.06
Gradient-boosted regression	0.68	15.4	2.9	1.24	0.57	1.19	0.36	1.10
Mean change of the same transition	0.39	22.3	1.5	1.12	0.10	1.07	0.26	2.26

Selection lots (1,314, two of five spatial folds) chose the form and fitted the baselines and the scale. Predicting no change gives a homes MAE of 16.5. Emissions: 115 benchmarked buildings.

5 Results

5.1 Existing conditions and planning calculations

The twin reproduces Manhattan's existing conditions (accounting checks).

- **Emissions.** Modelled emissions match 6,234 benchmarked buildings in a year not used for fitting (log–log $r = 0.807$, median error 31%, total 1.04 times metered).
- **Jobs** correlate with LEHD counts at tract level ($r = 0.843$) and block level ($r = 0.727$).
- **Homes.** Given observed use and floor area, homes on 295 lots developed in 2015–2025 total 1.04 times the recorded count ($r = 0.903$).
- **Environmental reviews.** Given two rezonings' final environmental impact statements (FEIS) and their own sites, floor area and uses (New York City Department of City Planning, 2017, 2021), PlanArena reproduces their projected homes at 0.86 and 1.00 times the review figures.

5.2 Effects of observed change

3,350 Manhattan lots changed use or floor area substantially between 2015 and 2025. Each was given its 2015 state and its observed new use and floor area, with coefficients from 2015 data. The test is conditional: it checks the engine given what was built, not whether the change would happen. The delta form, the unit-size rule and a scale factor were chosen on 1,314 lots in two of five spatial folds; results are for the other 2,036 (Table 4; Online Resource 1, Sect. S4).

- **Homes**: predicted change correlates with recorded change at $r = 0.79$, against 0.68 for gradient-boosted regression and 0.39 for the mean change of the same transition, both fitted on the selection lots with the same information. The mean absolute error is 12.5 homes per lot, against 16.5 for predicting no change.
- **Value**: $r = 0.74$, against 0.57, but the direction of the change (relative to the district trend) is right for only 60% of lots, against 88% for homes; **emissions** of 115 benchmarked buildings: $r = 0.60$, against 0.36.
- **Bias.** The engine overstates total homes added 1.47-fold (the regression baseline 1.24-fold). A scale factor fitted on the selection lots brings this to 1.35. Conversions to mixed use account for 70% of the excess; commercial-to-mixed conversions alone overstate by 28 homes per lot.
- **Jobs.** By census block, predicted job change does not correlate with the change in LEHD counts ($r = 0.05$): the engine has no skill for job change at this scale.
- **Learned correction** (Online Resource 1, Table S7). On the same held-out lots, a gradient-boosted model of the engine's error, fitted on the selection lots, lowered the mean absolute error from 12.5 to 10.6 homes per lot (total 1.47 to 1.32 times recorded) and lowered the error in value; for emissions it did not, and is not used. Conformal intervals with a nominal 90% coverage contained the recorded change for 88% of lots (homes), 85% (value) and 98% (emissions); the sensitivity ranges contained it for 66%, 19% and 38%.

Access, mix, green cover and runoff have no after-change data, and the Studio marks them as not validated.

5.3 Likelihood of redevelopment

Trained on 2015–2020 outcomes and tested on 2020–2025, the redevelopment model reaches AUC 0.855, against 0.802 for the best simple baseline and 0.833 for a graph neural network, with 26% redeveloped among its top 1% of lots against a base rate of 1.23%. Across 15 rolling origins (2016–2025; horizons of one, three and five years), its out-of-time AUC is 0.851–0.902, above the best baseline at every origin.

Calibration (Online Resource 1, Sect. S5). The model ranks lots better than it predicts how many redevelop.

Table 5 Gain in out-of-time AUC from adding a block, with 95% bootstrap intervals over 2 km tiles. Bold: the interval excludes zero.

City	C over E	Z over E+C	P over E+C	Z, conservative	P, conservative
Los Angeles	**+0.027 [0.020, 0.034]**	**+0.005 [0.001, 0.008]**	+0.002 [−0.002, 0.006]		
Dallas	**+0.021 [0.016, 0.026]**	**+0.003 [0.002, 0.005]**	**+0.002 [0.001, 0.003]**		
New York	**+0.023 [0.018, 0.029]**	**+0.004 [0.003, 0.007]**	–		
Milwaukee	−0.015 [−0.079, 0.048]	+0.029 [−0.031, 0.075]	–		
Phoenix	−0.066 [−0.139, 0.001]	**+0.049 [0.009, 0.091]**	**+0.054 [0.005, 0.100]**	**+0.055 [0.017, 0.088]**	**+0.041 [0.004, 0.073]**
San Diego	**+0.116 [0.053, 0.170]**	−0.003 [−0.022, 0.018]	**+0.021 [0.003, 0.039]**	−0.009 [−0.029, 0.015]	−0.011 [−0.033, 0.007]
Chicago	**+0.063 [0.049, 0.078]**	**+0.004 [0.001, 0.007]**	–	−0.001 [−0.005, 0.003]	–

Conservative: in each window, lots whose zoning or plan dates from the window's start or later are left out; possible where a current layer records those dates (Phoenix, San Diego, Chicago). Dallas's current layers carry no dates, so its zoning and plan gains may include post-decision information.

Table 6 What developed lots became: share named correctly, as one of 16 classes / one of nine groups.

City	Developed	No change	Zoning class	Plan class	Model (blocks)
Los Angeles	1,877	57% / 79%	70% / 87%	70% / 87%	75% / 89% (E+C+Z+P)
Dallas	3,886	37% / 40%	61% / 65%	80% / 90%	86% / 96% (E+C+Z+P)
New York	4,002	24% / 37%	49% / 65%	–	54% / 70% (E+Z)
Milwaukee	78	18% / 18%	37% / 77%	–	63% / 76% (E+Z)
Phoenix	1,835	0% / 35%	64% / 74%	47% / 62%	75% / 81% (E+C+Z+P)
San Diego	202	0% / 52%	14% / 74%	27% / 70%	64% / 75% (E+C+Z+P)

In Phoenix and San Diego development is a change of class (to another group, or to more homes), so "no change" is never right at class level there. Models are cross-fitted over 2 km tiles among the developed lots. Chicago's permits do not record the new use.

- **Totals shift with the cycle.** Observed over predicted redevelopments ranged from 0.07 to 1.70 across origins, and in the seven cities' test windows from 0.42 to 1.60.
- **Spread.** Calibration slopes were 0.59–0.95: the probabilities are too spread out.
- **Skill.** Against a constant forecast of the training-period rate, Brier skill was −0.97 to 0.15, negative at three of 15 origins.
- **Recalibration** on out-of-fold training probabilities corrects the spread (slopes 0.74–1.05) but not the totals, which still range 0.06–1.31 in Manhattan and 0.41–1.24 in the cities: the rate of development changes between windows in ways a model trained on earlier windows does not anticipate.

The probabilities are used to rank sites and to weight projections whose totals are compared with observed rates, not as forecasts of how many lots will redevelop.

5.4 What zoning and plans add, across seven cities

Existing conditions alone rank developed lots with out-of-time AUC 0.824–0.911 in six cities, and 0.571 in San Diego, whose one-year window recorded few changes (Table S14 in Online Resource 1; average precision in Table S16). Adding blocks:

- **Context** adds the most: +0.021 to +0.116, significant in five cities (Table 5, Figure 5). Phoenix is the exception: its land-use series starts in 2022, so recent change is unknown in its training window.
- **Zoning** carries its own signal. Without context, it raises AUC significantly in five of seven cities, and developed lots were 2.1–9.5 times as likely as others to have a zoning class outside their existing use group.
- **Overlap.** Most of that signal is shared with existing conditions and context: developed lots were also 3.5–19 times as likely to be vacant. Over context, zoning adds a significant gain in five of seven cities and the plan in three of four, at most +0.021 outside Phoenix.
- **Phoenix.** Zoning adds +0.049 and the plan +0.054, and 45% of developed lots were vacant.
- **Conservative variant.** The plan gain in San Diego and the zoning gain in Chicago do not survive it: those gains came from lots rezoned or re-planned during the study. Dallas's zoning and plan are undated current layers (the plan was adopted after the window's start), so its zoning and plan results may contain post-decision information.

The layers matter more for what developing lots become (Table 6).

- **Zoning class.** In all six cities with a destination, it names the new class correctly more often than the existing use does, by 12–64 percentage points. In Phoenix and San Diego, development is a change of class by definition, so the existing use is never right at class level.
- **Plan class.** In Dallas it is right for 80% of developed lots, against 61% for zoning.
- **Transfer.** Models fitted on other cities predict worse than the city's own in four of five cities (Table S15 in Online Resource 1).

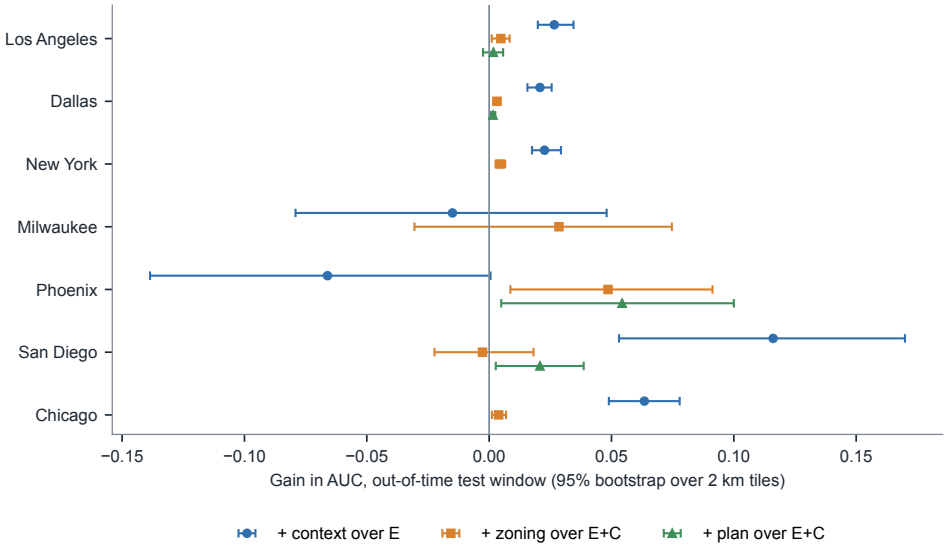


Fig. 5 Gain in out-of-time AUC from each block, with 95% tile-bootstrap intervals

Table 7 Homes added in the test window: projection over observed total (Spearman correlation across districts).

City	Observed	Zoning build-out	Plan build-out	Likely × zoning	Likely × plan	Likely × yield	Five-year trend
Los Angeles	22,853	193× (0.10)	189× (0.24)	0.80× (0.51)	0.81× (0.54)	0.33× (0.79)	0.59× (0.70)
Dallas	10,926	141× (0.00)	127× (0.34)	2.60× (−0.05)	1.59× (0.33)	0.45× (0.73)	1.25× (0.72)
New York	116,776	76× (0.34)	–	0.96× (0.55)	–	0.37× (0.83)	0.47× (0.81)
Milwaukee	3,798	115× (−0.23)	–	0.97× (−0.37)	–	0.12× (−0.28)	0.32× (0.51)

Observed: gross homes added in the window. Build-out (a proxy, not legal capacity): every lot at the homes per hectare that the 90th percentile of lots of its zoned (or planned) class reach in the city today. Likely: the build-out increment times the out-of-time probability of development (blocks E+C+Z+P where available). Yield: the probability times the mean homes added by developing lots of the same class and lot size in the training window. Trend: homes completed in the five years before the window, scaled to its length.

5.5 Housing projections

Projections and observations differ in kind: build-out is an end state, while the observed count is the gross number of homes added in a two- or four-year window (Table 7).

- **Ranking districts.** A five-year trend ranks districts better than build-out and likelihood × build-out in all four cities: Spearman 0.51 to 0.81, against −0.37 to 0.55. Likelihood × empirical yield (below) reaches −0.28 to 0.83, as well as the trend or better in three of four.
- **Build-out.** A proxy of zoning and plan build-out (each lot at the density that lots of its zoned or planned class reach today, not legal capacity) exceeds the homes added in the window 76- to 193-fold.
- **Totals.** Likelihood weighting brings totals to 0.80–2.60 times the observed. Using the homes that developing lots of the same class and size added in the training window instead of build-out gives 0.12–0.45: it ranks districts well but underestimates how many homes are built.
- **Manhattan rezonings** (Table 8, Figure 6). Capacity projected 8,423 homes on development sites, or 5,207 prorated to 2025; 561 were built. Probability × capacity gave 607, but the close total comes from errors that cancel: 0.84 and 0.87 times the observed in East Harlem and Inwood, and 44 times in SoHo/NoHo, where 3 homes were built. Soft sites redeveloped in the training years added 38%–102% of their capacity homes on average; scaling by that share gives 352 in total and a larger summed error across the three (471 homes, against 212), so the empirical yield did not improve these projections. The observed count lies within the 95% range of probability × capacity in two of three rezonings; the ranges are conditional on the model's probabilities and omit model error.

5.6 Optimizer and plausibility

Optimality (algorithmic verification).

- **Small areas.** In 20 tax blocks every plan with up to two changes was enumerated. The exact search returned the exhaustive optimum in all 20, with a bound never below it (median 0.6 s); first-order search found it in 14 of 15 of the blocks with a plan better than no change, and with exact re-scoring in all 15.
- **Areas the size of a lasso selection.** In 30 random areas of 28–411 parcels (27–323 eligible, a budget of 5%), the exact search certified the optimum in 29 within 120 s (median 3.2 s, at most 30 s). First-order search reached a

Table 8 Three Manhattan rezonings: homes on development sites by 2025, observed and projected.

Rezoning (adopted)	Sites	Built	Capacity	Pro-rated	Prob. × capacity (95%)	Prob. × yield (95%)
East Harlem (2017)	127	288	4,117	3,294	241 (82–497)	91 (0–444)
Inwood (2018)	47	270	2,859	1,334	234 (0–677)	127 (0–1,033)
SoHo/NoHo (2021)	77	3	1,447	579	132 (8–296)	134 (0–801)
All three		561	8,423	5,207	607	352
Sum of absolute errors			7,862	4,646	212	471

Capacity is full build-out by the review's build year; pro-rated scales it linearly to 2025. The probability comes from a redevelopment model trained only on outcomes known by the adoption year; yield is the mean share of capacity homes that soft sites redeveloped in those training years added. Ranges are conditional on the probabilities (Online Resource 1, Sect. S5).

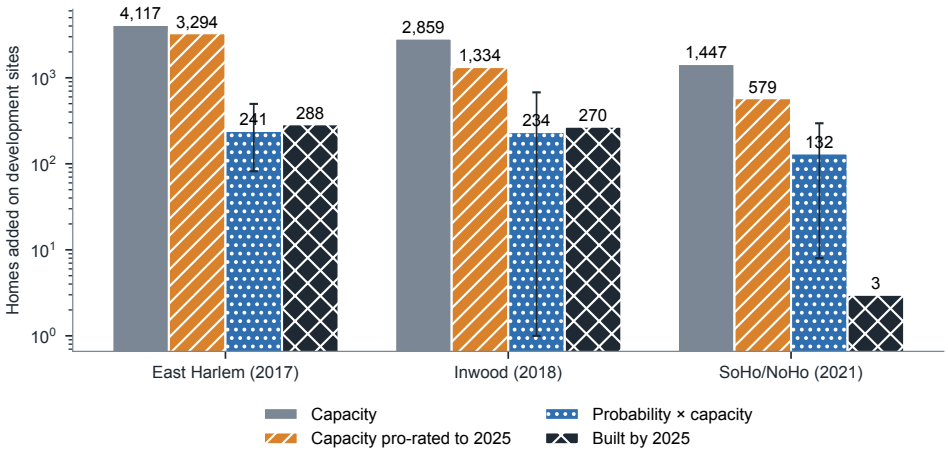


Fig. 6 Homes on development sites after three Manhattan rezonings: capacity, pro-rated capacity, probability × capacity (with its 95% range, conditional on the probabilities) and built by 2025 (log scale)

median 100.0% of the optimum and found it in 19 of 29 areas; with re-scoring, 100.0% and 21.

- **Districts.** With ten changes in each of the 12 districts and a limit of 3 hours, started from the best plan of the heuristics (Table 9), the exact search certified the optimum in nine districts, in 5–35 minutes (Table 10). In the other three the proven gap was 0.08% (district 104), 12.4% (district 101), 35.1% (district 105). Where it certified, the exact greedy plan was usually already optimal, which only the certificate shows; it found a better plan than every heuristic in one district. Where gaps remain, a few conversions add tens of thousands of residents, whose dilution of access across the district weakens the relaxation.

Trade-off front. In 10 random areas the exact front (pillar weights on a grid of step 0.25, and the balanced plan) had a median of 6 Pareto plans, with 100% of its points certified, in 143 s. On five small blocks enumerated exhaustively, 0 of its 20 plans were dominated by any plan, and it found 19% of the Pareto plans (weighted optima reach only the supported ones). The balanced plan raised every pillar by a median of 0.09 points at once: today's land use lies that far inside the frontier. 100% of the front stayed non-dominated under the low and high coefficient variants. NSGA-III covered a median 100.0% of the exhaustive front's hypervolume on 10 blocks with 24,576–327,680 plans each (Online Resource 1, Sect. S6).

GNN policy (held-out areas). On 80 areas in districts the network never saw, its plan (the network's most likely changes, scored exactly against the top first-order changes) reached a mean 98.8% of the certified optimum, and the optimum itself in 68% of areas, in 80 ms. First-order search reached 98.7% (75%) in 220 ms, and with exact re-scoring 99.6% (77%) in 945 ms. The network alone, without exact scoring, reached a median 97.8% (mean 85.7%) in 11 ms. Started from the network's plan, the exact search took a median 4.7 s, against 4.7 s from the first-order plan (20 areas): the network is a fast preview, not a faster route to the certificate.

Plausibility (Table 11). Plans were recommended by the fast search on the 2020 snapshot of every district and compared with 2025, with the same number of recommended parcels in each setting and one objective. An optimal plan for the score is no more a forecast than these.

- **Precedent.** Without realism layers, 17% of recommended changes had no precedent.

Table 9 Optimizer at district scale (12 Manhattan districts, up to ten changes): share of the best plan any tested method found (not a proven optimum).

Search	Median share of best	Districts best	Median seconds
surrogate top moves	91.0%	0	0.02
first order	94.1%	0	0.25
first order plus exact polish	95.7%	1	1.88
greedy exact	100.0%	12	36.15
simulated annealing	93.6%	0	10.65
random restarts 50	65.5%	0	1.31

Table 10 Exact search at district scale (ten changes): the best plan's score, the proven upper bound on any plan's score and the gap, after a time limit, started from the best heuristic plan.

District	Eligible	Best J	Bound	Gap	Certified	Seconds
101	645	6.664	7.487	12.4%	no	10,767
102	1,179	9.819	9.819	0.01%	yes	1,256
103	3,382	3.760	3.761	0.01%	yes	2,076
104	2,782	10.823	10.831	0.08%	no	10,812
105	2,179	4.222	5.704	35.1%	no	10,427
106	2,217	6.015	6.016	0.01%	yes	1,355
107	1,159	19.022	19.022	0.00%	yes	291
108	3,486	4.483	4.484	0.01%	yes	446
109	1,567	2.579	2.579	0.00%	yes	895
110	2,867	4.406	4.407	0.01%	yes	1,606
111	2,663	3.120	3.120	0.01%	yes	578
112	2,348	5.255	5.256	0.00%	yes	906

J: overall score minus the conversion penalty. Gap: (bound − best)/best.

- **Redevelopment.** Recommended parcels were redeveloped by 2025 at 11% without realism layers, 15% with the precedent filter and 21% with the likelihood layer, against 1.2% of all lots and 5.0% of the likeliest fifth.
- **Redevelopment is not implementation.** None of the recommended parcels became the recommended use.
- **The cost.** The objective fell by 3% with the precedent filter and 48% with the likelihood layer: the filter removes implausible conversions at little cost, while restricting candidates to likely sites trades much of the objective for plausibility.
- **The cost, exactly.** In 11 random areas, requiring the changed parcels' mean five-year redevelopment probability to reach the 50th, 80th or 95th percentile of the area's eligible parcels bound in 1, 3 and 6 areas and lowered the certified optimum by a mean 0.9%, 1.7% and 7.0% (at most 35%; 92% of the solves certified): a plausibility–benefit frontier.

5.7 Decision rules (illustrative)

The decision module has no observed counterpart: its outputs follow from assumed utilities, one seat per agent and the generated candidates. In the simulated scenario for 12 Manhattan districts:

- the nine rules never all agree, and a Condorcet winner exists in seven;
- the residents' referendum picks the council's plan in one district, and the council's winner is never the Nash bargaining solution;
- one insincere ballot could change the Schulze result in eight districts;
- the winner survives a median 68% of joint perturbations of coefficients and seats, and 58% of redrawn agent weights;
- the candidates matter: the nine rules pick a median of four distinct winners on the generated ballot, three on the agents' ideal plans alone and two on random feasible plans;
- with resident weights drawn independently for each block, the residents' winner is unchanged in a median 100% of draws.

These describe the simulated scenario, not public opinion. They make explicit what a hearing leaves implicit: resident and environmental interests are inputs to the decision, and a recommendation should be reported with its rule, its candidates and how it changes under others.

Table 11 Realism layers at equal counts: plans recommended on the 2020 snapshot of every Manhattan district, and what happened by 2025.

Optimizer setting	Parcels	No precedent	Redeveloped	Eligible redev.	Into rec. use	Objective
no realism layer	116	16.4%	11.2%	1.5%	0.0%	83.7
precedent filter (default)	117	0.0%	15.4%	1.5%	0.0%	81.3
precedent filter + likelihood layer	118	0.0%	21.2%	5.0%	0.0%	43.4

In each district every setting recommends the smallest number any setting recommended there. Eligible redev.: share of the setting's eligible parcels redeveloped (the rate of a random choice). Objective: the sum over districts of the overall score minus the conversion penalty. At the full budget the settings recommended 539, 543, 118 parcels.

6 Discussion

What a planner gets. For any proposed change, PlanArena shows:

- its effects on 11 indicators, with ranges;
- whether zoning and the plan allow it, and how likely it is;
- the optimal plan for an area, with a certificate, and what each pillar costs in the others;
- how the choice depends on the decision rule.

Each output has a stated test and type of evidence (Table 3), and untested indicators are labelled. For one system, this narrows the gap between claimed and demonstrated contributions of PSS and urban twins (Vonk et al., 2005; Azadi et al., 2025; Kruse et al., 2026).

Design guidance.

1. **Use each layer for the question it answers.** Once existing conditions and context were known, zoning and plans added little to ranking *where* development happened in these windows. They helped name *what* it became and bound *how much* a site can hold. This is consistent with rezoning following opportunity (Gabbe, 2018), upzoning's varying short-run effects (Freemark, 2020; Dong, 2024; Greenaway-McGrevy & Phillips, 2023) and partial plan implementation (Laurian et al., 2004; Berke et al., 2006).
2. **Report a trend first, and likelihood-weighted projections beside capacity.** Capacity overstates near-term housing by orders of magnitude. A trend ranked districts better than build-out-based projections in all four cities, and likelihood $\times$ yield as well as the trend in three of four. Close totals can hide large errors by area.
3. **Certify recommendations, and keep optimal and likely apart.** For areas of a few hundred parcels the optimum can be proven in seconds to minutes. First-order search found it in 19 of 29 areas, but only the certificate says which ones; it makes a recommendation checkable. But optimal for the score is not likely: realism layers change recommendations and lower the objective, and recommended sites that redevelop rarely become the recommended use.
4. **Measure trade-offs on the frontier.** Today's land use lay inside the frontier on every pillar at once, and the rates between pillars change along it; a single weighted score hides both.
5. **Report decisions with their rule.** No rule agreed with all the others in any district of the simulated scenario.

Limitations.

- **Impacts** are evaluated only in Manhattan and only given what was built; in the other nine cities they are unvalidated estimates from national or regional coefficients.
- **Indicators without skill or test.** Job change by block was tested and showed no skill; access, mix, green cover and runoff after a change have no observed test.
- **Calibration.** Redevelopment probabilities shift with the market cycle, so expected totals need checking against recent rates.
- **The cross-city test** uses short windows (one or two years) and city-specific outcome definitions, and Dallas's undated layers may include post-decision information.
- **Representation.** Zoning is reduced to use permissions and maximum floor-area ratios (no height, bulk, parking or bonus rules), access to straight-line distance, and uncertainty to sensitivity ranges, which are not prediction intervals; calibrated intervals exist only for homes and value, from Manhattan changes.
- **Distribution.** Indicators are averages over a scope. Who gains and who loses within it (by income, tenure or race), displacement and the effects on future residents are not measured.
- **Decisions.** Optimality is certified for the model's score, for areas of up to a few hundred parcels and nine of 12 districts; for the others the exact search proves a gap, not an optimum. The GNN policy has no guarantee of its own. The votes rest on assumed utilities and one seat per interest, not on elicited preferences or representation.
- **Not studied.** Usability with planners is not evaluated, and causal effects of zoning or plans are not estimated.

7 Conclusion

Land-use and location decisions have multidimensional, spatially interacting and uncertain effects, yet they are decided at hearings where those effects are summarised in a few indicators and residents do not vote. PlanArena, a land-use optimizer, measures the effects of any parcel-level change on social, economic and environmental indicators, with what is permitted, planned and likely, finds and certifies the optimal plan of an area, measures the trade-offs between pillars on the exact frontier, and turns stakeholder interests into votes. Machine learning corrects the engine's yields with calibrated intervals and proposes optimal plans at once. It runs on official data for ten large US cities at interactive speed, and each component has a stated test, empirical where data allow. The tests also shape the design:

- once existing conditions and context are known, regulatory layers mainly inform what development becomes and how much it could hold;
- a five-year trend ranked districts better than build-out projections, and likelihood-weighted totals were far closer to what was built than capacity, partly through errors that cancel;
- an exact search proves the optimal plan of an area in seconds where heuristics fall short, but optimal is not likely: recommendations change when they must be plausible, at a measured cost, and the chosen plan changes with the rule and the candidates.

Code, data manifest and results are released with the paper, so the system and its tests extend to other cities.

Statements and Declarations

Competing interests. The author has no relevant financial or non-financial interests to disclose.

Funding. No funding was received for conducting this study.

Ethics approval and consent. Not applicable. The study uses public administrative records; it involves no human participants or animals. Owner, mailing and taxpayer fields in assessor records were never requested or read.

Data availability. All inputs are public and are downloaded by script; a manifest records each input's URL, retrieval date and SHA-256, and the result file behind every number, the figures and each city's ingestion report are in the repository below. The datasets are cited in the text and listed in the references.

Code availability. The code is released under the MIT licence at `https://anonymous.4open.science/r/planarena-cus` (anonymised for review); `make reproduce` rebuilds the data and every result. The online Studio and game are at `https://planarena-manhattan.onrender.com` and `https://planarena-manhattan.onrender.com/play`.

Author contributions. Given on the title page.

A Digitising plans published as PDF maps

Philadelphia2035's land-use recommendations exist only as "Proposed Land Use" maps in 17 district plans covering 18 districts. Each lot is a filled vector shape. The maps are converted in six steps:

1. **Legend.** Swatch colours and labels are read.
2. **Colour matching.** Map colours are matched one to one to legend colours by CIELAB ΔE with a Hungarian assignment, refusing pairs with $\Delta E > 45$.
3. **Visibility.** Only shapes the rendered page shows are kept, since maps are often clipped to the district, and only the largest map frame on the spread.
4. **Registration.** A similarity transform is fitted. The rendered footprint is matched as an image to the district's parcels and outline (FFT translation, Dice score), then refined by iterative closest point (Reddy & Chatterji, 1996; Besl & McKay, 1992; Umeyama, 1991).
5. **Classification.** Each parcel takes the category of the smallest shape containing its interior point.
6. **Checks.** A map is used only if the median parcel-to-shape distance is at most 3.0 m (10 ft), and its plan agrees with existing use clearly more often than the same map shifted 91 m (300 ft).

Results:

- **Coverage.** Maps from 15 of 17 plans passed, covering 16 districts and 83% of parcels.
- **Registration.** Median errors were 0.3–2.2 m (median 1.0 m).
- **Shifted-map control.** The digitised plans agree with existing use on 60%–90% of parcels, against 42%–66% for shifted maps.
- **Independent reading.** Each parcel's category was also read from the colour of the rendered page around its point, matched to the legend's rendered colours. On 451,691 parcels read both ways, the two readings agree on 99.8% (99.4%–100.0% by district). Both use the same registration, which the shifted-map control checks. A random sample of 150 parcels, with page positions and map crops, is provided for a check by eye against the PDFs.
- **Failures.** The maps of South and West Park also draw neighbouring districts and failed registration.

Earlier work addresses scanned raster maps (Chiang et al., 2014; Luft & Schiewe, 2021; Milleville et al., 2022); vector exports allow shapes to be read directly.

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